The Continuous-Coefficient Jensen Equation: How Real Mixing Weights Retire Regularity

Double Blind No Name

June 11, 2026

Abstract

Jensen's inequality is a workhorse of analysis: for every concave function G and random variable ξ on an interval, $\mathbb{E}[G(\xi)] \leq G(\mathbb{E}[\xi])$. What if equality is forced for every two-point distribution — i.e., $p\,G(u_1)+(1-p)\,G(u_2) = G(p\,u_1+(1-p)\,u_2)$ for all u_1, u_2 in the interval and all weights $p \in [0, 1]$? Must G be affine?

Yes. The proof is one line: substitute $u_1 = M$, $u_2 = 0$, $p = v/M$ and read $G(v)$ off the straight line through $(0, G(0))$ and $(M, G(M))$. No regularity hypothesis on G — continuity, measurability, monotonicity, or boundedness — is needed.

The result contrasts sharply with what happens when the weights p are restricted to rationals. The $p = \frac{1}{2}$ specialization is equivalent to Cauchy's additive equation, which admits wildly pathological non-affine solutions (Hamel, 1905) without some regularity hypothesis. Classical analysis from Cauchy (1821) to Kestelman (1947) cataloged the minimal tameness conditions — continuity, measurability, boundedness on a set of positive measure — that rule the pathology out. These conditions became standard baggage in functional-equation arguments, carried reflexively even where the full continuous-coefficient equation is in play and no such baggage is needed.

This expository article makes the asymmetry explicit. We catalog which classical regularity hypotheses become vestigial under the continuous-coefficient equation, explain structurally why (the Hamel pathology lives at irrational p, exactly where the continuous-coefficient equation tests G), and then give a diagnostic that separates the saturated-Jensen settings in which these hypotheses are vestigial from the more common settings in which they remain load-bearing. The same five hypotheses appear on both sides of the line: in the axiomatic characterization of Shannon entropy they are indispensable, while in a calibration computation on an atomless space they are not, and we explain what distinguishes the two.

Keywords: Jensen equation, Cauchy equation, Hamel basis, functional equation, regularity hypothesis

MSC 2020: 39B22, 39B05

1 Introduction.

One of the most serviceable inequalities in mathematics—Jensen's inequality, $\mathbb{E}[G(\xi)] \le G(\mathbb{E}[\xi])$ for concave G—raises an immediate follow-up: when does equality hold for *every* two-point distribution supported on a given interval? The equation expressing universal two-point equality,

$$p\,G(u_1) + (1-p)\,G(u_2) \;=\; G\big(p\,u_1 + (1-p)\,u_2\big) \qquad (u_1, u_2 \in I,\ p \in [0,1]), \tag{$\star$}$$

imposed on a function $G\colon I \to \mathbb{R}$ on a real interval I, has at least three forms in classical analysis. These forms have been steadily conflated for over a century, with consequences for both the foundational theory and the modern applied literature where the equation surfaces.

1.1 Three forms of the equation.

The *discrete-coefficient* Jensen equation,

$$G\big(\tfrac{u_1+u_2}{2}\big) \;=\; \tfrac{G(u_1)+G(u_2)}{2}, \tag{J_2}$$

is the $p = \frac{1}{2}$ specialization of ($\star$). Setting $f(x) := G(x) - G(0)$, the equation (J_2) is equivalent on a translate of I to Cauchy's additive equation $f(x+y) = f(x) + f(y)$. Cauchy's equation inherits a full apparatus of pathological solutions: without measurability, monotonicity, boundedness on a set of positive Lebesgue measure, or another regularity hypothesis, the equation admits non-affine solutions constructed via a *Hamel basis*—a basis of $\mathbb{R}$ as a $\mathbb{Q}$-vector space, whose existence requires the axiom of choice. The classical pathological-solution apparatus is the work of Hamel [1]; the regularity hypotheses sufficient to recover affineness were cataloged by Cauchy [2], Darboux [3, 4], Sierpiński [5], Kormes [6], Ostrowski [7], and Kestelman [8], with Steinhaus's difference-set theorem [9] supplying the tool behind the modern proofs of the measure-theoretic entries; Reem [10] surveys the catalog.

The *rational-coefficient* Jensen equation,

$$p\,G(u_1) + (1-p)\,G(u_2) \;=\; G\big(p\,u_1 + (1-p)\,u_2\big) \qquad (u_1, u_2 \in I, \quad p \in [0,1] \cap \mathbb{Q}), \tag{$J_{\mathbb{Q}}$}$$

strengthens (J_2) but does not escape its pathology. Cauchy additivity entails $\mathbb{Q}$-homogeneity—$f(qx) = qf(x)$ for every $q \in \mathbb{Q}$ and $x \in \mathbb{R}$—by an elementary induction on positive integers, sign reversal for negatives, and the inverse-of-multiplication argument for $1/n$. Combining additivity with $\mathbb{Q}$-homogeneity returns ($J_{\mathbb{Q}}$) for f, hence for G. So ($J_{\mathbb{Q}}$), (J_2), and Cauchy's equation share the same solution class up to constants, and all three inherit the Hamel-basis pathology.

The *continuous-coefficient* form ($\star$), by contrast, is fundamentally different. The strengthening from $\mathbb{Q}$ to $\mathbb{R}$ in the range of p is *genuine*. As we will see, ($\star$) does not admit any Hamel pathology: the pathology cannot survive at irrational p, and the continuous-coefficient equation tests G at every irrational. This is the subject of the paper.

1.2 The main result.

Theorem 1. *Let $M > 0$ and let $G\colon [0, M] \to \mathbb{R}$ satisfy $(\star)$ for every $u_1, u_2 \in [0, M]$ and every $p \in [0, 1]$. Then G is affine on $[0, M]$:*

$$G(v) \;=\; \frac{G(M) - G(0)}{M}\, v + G(0), \quad v \in [0, M].$$

No regularity hypothesis on G (continuity, measurability, monotonicity, boundedness) is required.

The proof is one line: a substitution into $(\star)$ that pins $G(v)$ on the straight line through the endpoints $(0, G(0))$ and $(M, G(M))$. We give it in Section 2 and use the remainder of the paper to articulate what it means.

1.3 The historical detour and the structural moral.

Theorem 1 is elementary. The underlying substitution technique is folklore in functional equations: Aczél's [11, §2.1.4] contains a dyadic precursor proved (with a continuity hypothesis) by the same kind of substitution. The precise statement of Theorem 1 — continuous coefficients $p \in [0, 1]$, no regularity hypothesis on G — does not appear in this exact form in the standard references the author consulted; it is recorded here because the recurrence pattern of Section 5 needs it as a separable lemma. The contribution of the present paper is in three parts:

- *The explicit dictionary* (Section 3) of classical regularity hypotheses on G — continuity, measurability, monotonicity, boundedness on a set of positive measure, and boundedness on the interval — each of which is required to force affineness for (J_2) but becomes *vestigial* under $(\star)$, together with the structural mechanism (the Hamel pathology lives at irrational p, exactly where $(\star)$'s continuous coefficient closes the door).

- *The strict-minimum hypothesis* (Section 4.1)—the equation $(\star)$ at a single endpoint configuration already suffices.

- *The diagnostic* (Section 5) for distinguishing saturated-Jensen derivations that genuinely produce the continuous-coefficient equation — where the dictionary's regularity hypotheses are vestigial — from those that produce a rational-coefficient relative, an additive function into a bounded codomain, a Cauchy relative with a non-affine target, or no real-valued unknown at all (the zeroth gate). In the latter cases the same hypotheses — or an existence axiom prior to them — are load-bearing; the misclassification runs in both directions, and we give examples of each.

Stylistically, this article follows the tradition of expository programs that concentrate on the part of Jensen-equation theory which does not require regularity assumptions on the function; Kuczma's [12] monograph is the canonical modern reference. The substitution that closes $(\star)$ is one of the simplest observations in that tradition; what justifies writing it down is the dictionary, the structural mechanism, and the survey of applied recurrences.

The classical detour through Hamel and the Polish school answered a perfectly natural question—*what regularity hypothesis suffices for* (J_2)⇒*affine?*—but a *different* question from the one that matters for many modern applications. The applied derivation does not produce (J_2) or $(J_{\mathbb{Q}})$; it produces $(\star)$ as a saturated identity, and the one-line substitution of Section 2 is the right closure tool. The dictionary of Section 3 makes explicit which classical regularity hypotheses become vestigial in this transition.

The pattern — a continuum built into the model retiring a classical hypothesis — has a distinguished ancestor. In exchange economies with an atomless continuum of traders, Aumann showed that convexity of preferences becomes unnecessary [13], Lyapunov-type convexity supplying for free what the hypothesis used to buy. The present article records the same phenomenon one level down, at the level of a single functional equation on an interval.

Terminology. Throughout the paper we refer to the substitution $u_1 = M$, $u_2 = 0$, $p = v/M$ in $(\star)$ — by which $G(v)$ is read off the straight line through the endpoints $(0, G(0))$ and $(M, G(M))$ — as the *endpoint substitution*. The substitution technique is standard folklore in the functional-equations literature (Aczél's [11, §2.1.4] proves a dyadic version); the descriptive label is our convenience, adopted for brevity in repeated reference rather than as a claim of established terminology.

2 The result and its proof.

Proof of Theorem 1. Fix $v \in [0, M]$. Set

$$u_1 := M, \qquad u_2 := 0, \qquad p := \frac{v}{M}.$$

The constraints of $(\star)$ are satisfied: $u_1 = M \in [0, M]$, $u_2 = 0 \in [0, M]$, and $p = v/M \in [0, 1]$ since $v \in [0, M]$. Substituting into $(\star)$,

$$\frac{v}{M}\, G(M) + \Big(1 - \frac{v}{M}\Big)\, G(0) \;=\; G\Big(\frac{v}{M} \cdot M + \Big(1 - \frac{v}{M}\Big) \cdot 0\Big) \;=\; G(v).$$

Rearranging yields $G(v) = G(0) + \big(G(M) - G(0)\big)\, v/M$. □

The endpoint substitution pins $G(v)$ on the straight line from $(0, G(0))$ to $(M, G(M))$ for every $v \in [0, M]$. Since this determines $G(v)$ at every v, the function G is fully determined by its values at the two endpoints — it is exactly the affine function joining them.

Corollary 2. *Under the hypotheses of Theorem 1, G is in particular continuous, monotone, locally Lipschitz, absolutely continuous, and Lebesgue measurable on* $[0, M]$.

Proof. Every affine function $G(v) = av + b$ on a bounded interval is differentiable with constant derivative $G' = a$, hence continuous; non-decreasing if $a \geq 0$ and non-increasing if $a \leq 0$; Lipschitz with constant $|a|$; absolutely continuous; and Borel-measurable, hence Lebesgue-measurable. □

The order of inference matters and is worth stating explicitly. Under $(\star)$, the regularity properties of G are the *conclusion*, not the hypothesis. Each regularity hypothesis one might be tempted to assume—continuity, monotonicity, boundedness, measurability—is automatically satisfied by any solution of $(\star)$. The continuous-coefficient equation supplies its own regularity from within; no external regularity hypothesis is needed.

This is what makes the continuous-coefficient case structurally different from the discrete-coefficient case (J_2), where some regularity hypothesis *is* required to rule out the Hamel pathology. We now examine the difference in detail.

3 The dictionary of vestigial regularity hypotheses.

3.1 The dictionary.

We place side-by-side the discrete-coefficient equation (J_2) and the continuous-coefficient equation $(\star)$, and ask which regularity hypotheses on G are *required* to force affineness for each. Table 1 lists these hypotheses side-by-side, and Figure 1 draws the comparison as a load diagram: five classical hypotheses, each individually strong enough to carry the conclusion for (J_2), and the same conclusion for $(\star)$ carried by the bare equation, every strut removed. The caption of Figure 1 records the standard historical attribution of each strut. (For a modern textbook treatment of the regularity catalog for (J_2) we refer the reader to Kuczma [12, §13.2].)

The ground line is the point of the figure. With **no hypothesis at all**, the beam falls for (J_2): there exists a non-affine, non-measurable, unbounded additive function on $\mathbb{R}$—the classical Hamel-basis pathology [1]—that supplies a non-affine solution of (J_2). Conversely, **no hypothesis is needed** for $(\star)$: Theorem 1's endpoint substitution closes the proof at the level of the bare functional equation.

The five struts tell a complementary story. For (J_2), *some* strut is required; otherwise the Hamel pathology survives. For $(\star)$, *none* is required—and yet each is automatically satisfied by any solution (Corollary 2). Each hypothesis is, in the dictionary's terminology, *vestigial*: classically required for (J_2), classically expected for $(\star)$, but in fact unnecessary for $(\star)$.

3.2 The structural mechanism.

Why does the Hamel pathology fail to apply to $(\star)$? The mechanism is direct and illuminating.

A Hamel-basis pathological solution G of Cauchy's equation on $\mathbb{R}$ is $\mathbb{Q}$-linear: $G(0) = 0$ and $G(qx) = q\,G(x)$ for every $q \in \mathbb{Q}$ and $x \in \mathbb{R}$. By design, G is *not* $\mathbb{R}$-linear: there exists an irrational $r \in \mathbb{R}$ and an $x \in \mathbb{R}$ such that $G(rx) \neq r\,G(x)$.

The straight-line identity in $(\star)$ at the configuration $u_1 = M$, $u_2 = 0$ reads

$$p\,G(M) + (1-p)\,G(0) \;=\; G(p\,M). \qquad (\star_0)$$

For a Hamel-pathological G (so $G(0) = 0$), $(\star_0)$ becomes $p\,G(M) = G(p\,M)$—exactly the $\mathbb{R}$-linearity assertion for G at the pair (M, p). By $\mathbb{Q}$-linearity of G, this holds for

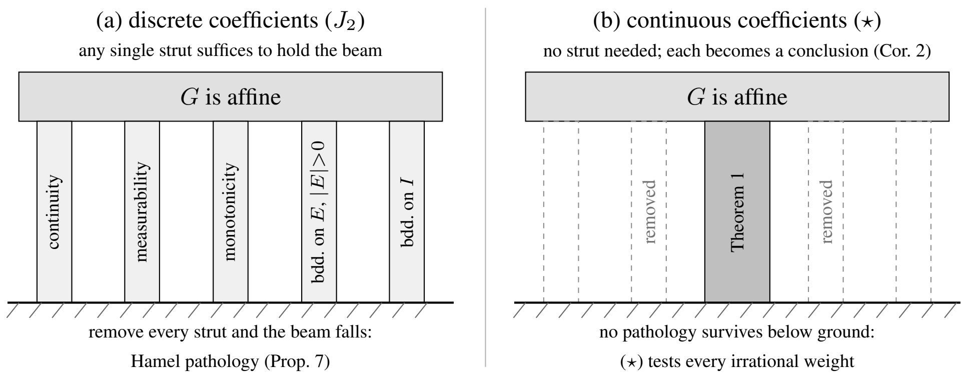

Figure 1: The dictionary of vestigial regularity hypotheses, drawn as a load diagram. (a) Under the discrete-coefficient equation (J_2), the conclusion "G is affine" must be carried by a regularity strut, and any single one suffices: continuity (Cauchy [2]), measurability (Sierpiński [5]), monotonicity (Darboux [3]), boundedness on a set of positive measure (Kormes [6]; one-sided bounds suffice, Ostrowski [7], Kestelman [8]), or boundedness on the interval (Darboux [4]). Remove them all and the beam falls: the Hamel-basis pathology [1] supplies a non-affine solution (Proposition 7). Steinhaus's difference-set theorem [9] powers the modern proofs of the two measure-theoretic struts but is a tool, not a strut; for the historical catalog see Reem [10] and Kuczma [12, §13.2]. (b) Under the continuous-coefficient equation ($\star$) the bare equation carries the load (Theorem 1); every strut becomes a conclusion rather than a hypothesis (Corollary 2), and no pathology survives below ground, because ($\star$) tests every irrational weight.

every $p \in \mathbb{Q}$. By the failure of $\mathbb{R}$-linearity, it *fails* at some irrational p. So the Hamel-pathological G satisfies ($J_{\mathbb{Q}}$) (every rational coefficient is OK) but violates ($\star$) at every irrational p where the $\mathbb{R}$-linearity defect appears.

The structural point is one sentence:

> **The Hamel pathology lives at irrational p; the continuous-coefficient equation ($\star$) forecloses on it there.**

Figure 2 draws both halves of that sentence.

The classical program from Cauchy (1821) through Kestelman (1947) can be read, in hindsight, as an investigation of what regularity hypothesis can substitute for the *unavailable* constraint at irrational p, once it became clear — after Hamel's 1905 construction — that the rational-coefficient version of the equation does not by itself force affineness. With the continuous-coefficient version, the irrational-coefficient constraint is *available*, and the entire dictionary collapses.

Hypothesis on G	**For** (J_2)⇒**affine?**	**For** $(\star)$⇒**affine?**
Continuity on I: G is continuous at every $v \in I$.	Yes, suffices (Cauchy [2]).	**No**—it is the *conclusion* (Corollary 2).
Measurability on I: G is Lebesgue- or Borel-measurable on I.	Yes, suffices (Sierpiński [5]).	**No**—same.
Monotonicity on I: G is non-decreasing or non-increasing on I.	Yes, suffices (Darboux [3]).	**No**—same.
Boundedness on a set of positive measure: G is bounded on some Lebesgue-measurable $E \subseteq I$ with $\lvert E\rvert > 0$.	Yes, suffices (Kormes [6]; bounded on one side suffices, Ostrowski [7], Kestelman [8]).	**No**—same.
Boundedness on I: G is bounded on all of I.	Yes, suffices (Darboux [4]).	**No**—same.
None: no regularity hypothesis at all.	**Insufficient**—Hamel-basis pathology [1].	**Sufficient**—Theorem 1.

Table 1: Regularity hypotheses on G. Column 2 records which hypotheses are classically required to force affineness for the discrete-coefficient equation (J_2). Column 3 records their status under the continuous-coefficient equation $(\star)$. Steinhaus's difference-set theorem [9] powers the modern proofs of the two measure-theoretic rows but is itself a tool rather than a regularity result; for the historical catalog see Reem [10] and Kuczma [12, §13.2].

4 Variants and limits.

4.1 The strict-minimum hypothesis.

The proof of Theorem 1 uses $(\star)$ at exactly one configuration: $u_1 = M$, $u_2 = 0$, with $p \in [0, 1]$ ranging freely. The full force of $(\star)$—for every pair (u_1, u_2)—is not consumed. We record the strict minimum.

Theorem 3. *Let $M > 0$ and let $G\colon [0, M] \to \mathbb{R}$ satisfy*

$$p\,G(M) + (1-p)\,G(0) \;=\; G(p\,M) \qquad \textit{for all } p \in [0,1].$$

Then $G(v) = G(0) + (G(M) - G(0))\,v/M$ on $[0, M]$.

Proof. Set $p := v/M$; then $p \in [0, 1]$ since $v \in [0, M]$. The hypothesis gives

$$p\,G(M) + (1-p)\,G(0) \;=\; G(p\,M) \;=\; G(v),$$

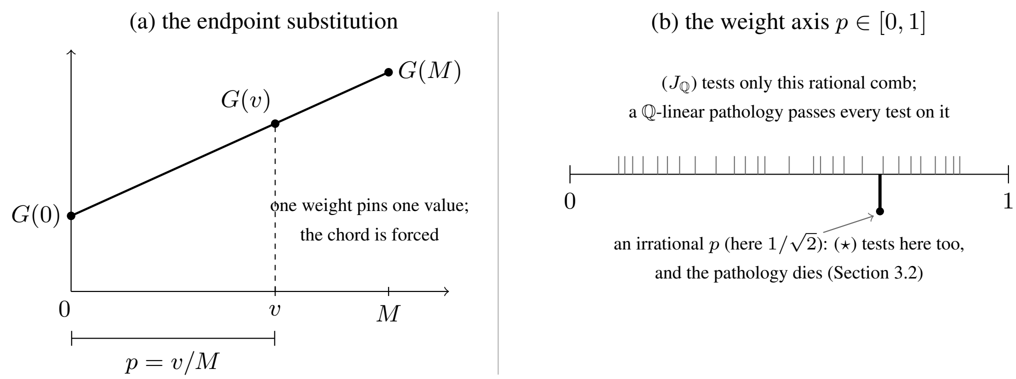

Figure 2: The mechanism. (a) The endpoint substitution that proves Theorem 1: at the configuration $u_1 = M$, $u_2 = 0$, the single weight $p = v/M$ pins $G(v)$ to the chord through $(0, G(0))$ and $(M, G(M))$ — one weight per point, the chord forced everywhere. (b) The weight axis on which the two equations live. The rational comb is where ($J_{\mathbb{Q}}$) tests the identity, and a $\mathbb{Q}$-linear Hamel pathology passes every test on the comb. The continuous-coefficient equation ($\star$) also tests the irrational weights, where the $\mathbb{R}$-linearity defect must show itself, and forecloses on the pathology there.

so $G(v) = G(0) + p\big(G(M) - G(0)\big) = G(0) + \big(G(M) - G(0)\big)v/M$. $\square$

Theorem 3 is *in principle* weaker than Theorem 1: the one-configuration hypothesis does not a priori imply the full ($\star$). Under the conclusion (affineness) both hold; in practice an author who derives ($\star$) from a richer setup (e.g., a Jensen-equality identity in a Bayes-risk computation; see Section 5) typically obtains ($\star$) in full force. The narrower Theorem 3 is the right reference when one wants to know exactly how much of the equation the proof actually consumes.

Two further variations sharpen the picture of how little input the conclusion needs, and of what survives when the full hypothesis is cut down.

Remark 4 (Irrational weights alone suffice)**.** The weight set in Theorem 1 need not contain a single rational. Suppose ($\star$) holds merely for every *irrational* $p \in (0,1)$, and let $L(v) := G(0) + \big(G(M) - G(0)\big)v/M$ denote the affine interpolant of the endpoint values. If v/M is irrational, the endpoint substitution at $p = v/M$ gives $G(v) = L(v)$ directly. If v/M is rational with $0 < v < M$, fix the irrational weight $p = \frac{\sqrt{2}}{2}$ and choose $\varepsilon > 0$ with $\varepsilon/M = \frac{\sqrt{3}}{n}$ for an integer n large enough that $u_1 := v + (1-p)\varepsilon$ and $u_2 := v - p\varepsilon$ lie in $[0, M]$. Then u_1/M and u_2/M are irrational, being a rational plus the irrational numbers $\frac{\sqrt{3}}{n} - \frac{\sqrt{6}}{2n}$ and $-\frac{\sqrt{6}}{2n}$, respectively. Since $p\,u_1 + (1-p)\,u_2 = v$, the hypothesis at the triple (u_1, u_2, p) together with the already-established values $G(u_i) = L(u_i)$ gives $G(v) = p\,L(u_1) + (1-p)\,L(u_2) = L(v)$, because L is affine. The Hamel pathology lives at irrational weights (Section 3.2); it is fitting that irrational weights alone suffice to exclude it.

Corollary 5 (Piecewise saturation)**.** *Let $0 = m_0 < m_1 < \cdots < m_k = M$ and suppose $G\colon [0, M] \to \mathbb{R}$ satisfies $(\star)$ separately on each cell $[m_{i-1}, m_i]$, that is, for all u_1, u_2 in a common cell and all $p \in [0, 1]$. Then G is affine on each cell, hence piecewise affine on $[0, M]$ and — being a single function whose affine pieces share their values at the knots — automatically continuous, with no regularity hypothesis. Global affineness does not follow: the slopes may differ across knots.*

Proof. Apply Theorem 1 on each cell after the affine change of variable $v \mapsto (v - m_{i-1})/(m_i - m_{i-1})$. Continuity at each knot m_i holds because the affine formulas on the two adjacent cells both evaluate there to the same number, $G(m_i)$. □

The corollary is the form in which Theorem 1 is consumed by the calibration setting of Section 5.3: a surrogate score on $[0, 1]$ that aggregates *exactly* on each of $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$ is necessarily piecewise affine — the tent shape — with nothing assumed at the kink or anywhere else.

4.2 Convex domains in higher dimensions.

The endpoint substitution extends to any convex subset of a real vector space.

Theorem 6. *Let V be a real vector space (not restricted to finite dimensions), let $C \subseteq V$ be a convex set, and let $G\colon C \to \mathbb{R}$ satisfy*

$$p\, G(x_1) + (1 - p)\, G(x_2) \;=\; G\big(p\, x_1 + (1 - p)\, x_2\big) \qquad (x_1, x_2 \in C,\ p \in [0, 1]).$$

Then there is a linear functional $a\colon \operatorname{span}(C - C) \to \mathbb{R}$ on the parallel subspace $W := \operatorname{span}(C - C)$ (the subspace parallel to the affine hull of C) and a constant $b \in \mathbb{R}$ such that $G(x) = a(x - x_0) + b$ on C for any fixed $x_0 \in C$. Equivalently, G is affine on C.

Proof. Fix $x_0 \in C$, let $D := C - x_0$ (a convex set containing 0), and set $A(v) := G(x_0 + v) - G(x_0)$ for $v \in D$, so that $A(0) = 0$.

(i) Real homogeneity on D. For $v \in D$ and $p \in [0, 1]$, the hypothesis at $x_1 = x_0 + v$, $x_2 = x_0$ gives $A(pv) = p\, A(v)$ — the endpoint substitution of Theorem 1, now run along the segment $[x_0, x_0 + v]$. This is the step where the continuum of weights acts; everything after it is bookkeeping in which $\mathbb{Q}$ plays no role.

(ii) Additivity where defined. For $v, w \in D$ the hypothesis at $x_1 = x_0 + v$, $x_2 = x_0 + w$, $p = \frac{1}{2}$ gives $A\big(\frac{v+w}{2}\big) = \frac{1}{2}\big(A(v) + A(w)\big)$; if moreover $v + w \in D$, then (i) at $p = \frac{1}{2}$ turns the left-hand side into $\frac{1}{2}A(v + w)$, whence $A(v + w) = A(v) + A(w)$ whenever $v, w, v + w \in D$.

(iii) Extension to a linear functional. On the cone $D^* := \bigcup_{\lambda > 0} \lambda D$ (which is a convex cone generated by D) define $\hat{A}(\lambda v) := \lambda\, A(v)$ for $v \in D$, $\lambda > 0$. This is well defined: if $\lambda v = \mu w$ with $v, w \in D$, scale by $t > 0$ small enough that $t\lambda, t\mu \le 1$; then $t\lambda v = t\mu w \in D$ (since $0 \in D$ and D is convex, $sD \subseteq D$ for $s \in [0, 1]$), and two applications of (i) give $\lambda A(v) = \mu A(w)$. The same scaling device upgrades (ii) to additivity of $\hat{A}$ on all of D^*, and (i) to $\hat{A}(cu) = c\, \hat{A}(u)$ for every real $c > 0$ and $u \in D^*$. Finally, $\operatorname{span}(C - C) = D^* - D^*$; for $u = v - w$ with $v, w \in D^*$ set $a(u) := \hat{A}(v) - \hat{A}(w)$. Additivity of $\hat{A}$ makes a well defined and additive, and

positive homogeneity extends to full $\mathbb{R}$-homogeneity via $a(-u) = -a(u)$, so a is linear. Then $G(x) = a(x - x_0) + G(x_0)$ on C, and any function of this form satisfies the hypothesis. $\square$

The structurally essential step is the one-dimensional endpoint substitution of Theorem 1. The higher-dimensional lifting is bookkeeping (compare the regularity-laden treatments of the Jensen equation on convex domains in Kuczma [12, §13.2] and Aczél–Dhombres [14]).

4.3 The rational-coefficient version retains the pathology.

The strengthening from rational to continuous coefficients in ($\star$) is doing genuine work: the rational-coefficient version ($J_{\mathbb{Q}}$) retains the Hamel pathology.

Proposition 7 (Folklore; cf. [12, §§5.2 and 13.2])**.** *There exists $G\colon [0,1] \to \mathbb{R}$ satisfying ($J_{\mathbb{Q}}$) (hence (J_2)) on $[0,1]$ that is not affine.*

Construction. Choose a Hamel basis H_0 of $\mathbb{R}$ over $\mathbb{Q}$ containing 1; such a basis exists by Zorn's lemma (which is equivalent to the Axiom of Choice; without it, as shown by Solovay [51], one cannot prove the existence of non-affine solutions to the Cauchy or Jensen equations). Pick any $h' \in H_0 \setminus \{1\}$; since two distinct rationals are $\mathbb{Q}$-linearly dependent, h' is irrational. Choose a rational $q > 0$ with $h := qh' \in (0,1)$, and set $H := (H_0 \setminus \{h'\}) \cup \{h\}$. Replacing a basis element by a nonzero rational multiple of itself preserves both $\mathbb{Q}$-linear independence and spanning, so H is again a Hamel basis, and it contains 1 together with the irrational $h \in (0,1)$.[1] Define a $\mathbb{Q}$-linear map $\ell\colon \mathbb{R} \to \mathbb{R}$ by setting $\ell(1) = 0$, $\ell(h) = 1$, and extending $\mathbb{Q}$-linearly on the remaining basis elements.

By construction, ℓ is $\mathbb{Q}$-linear: $\ell(x + y) = \ell(x) + \ell(y)$ (Cauchy's equation) and $\ell(qx) = q\,\ell(x)$ for all $q \in \mathbb{Q}$ ($\mathbb{Q}$-homogeneity). Set $G := \ell|_{[0,1]}$; then G satisfies ($J_{\mathbb{Q}}$) on $[0,1]$ because $\mathbb{Q}$-rational convex combinations of points of $[0,1]$ remain in $[0,1]$ and ℓ respects $\mathbb{Q}$-linear combinations.

Explicit witness that G is not affine: $G(h) = \ell(h) = 1 \neq 0$, while $G(q) = \ell(q) = q\,\ell(1) = 0$ for every rational $q \in [0,1]$. Because $\mathbb{Q} \cap [0,1]$ is dense in $[0,1]$, this pathological G oscillates wildly: its graph is a dense subset of $[0,1] \times \mathbb{R}$, violating every single regularity condition of Table 1 (or Figure 1) simultaneously. Any affine map $A\colon [0,1] \to \mathbb{R}$ agreeing with G on $\mathbb{Q} \cap [0,1]$ would satisfy $A(0) = 0$ and $A(q) = 0$ for all rational q—forcing $A \equiv 0$, contradicting $G(h) = 1$. $\square$

Proposition 7 sharpens the dictionary: ($J_{\mathbb{Q}}$) on $[0,1]$ admits non-affine solutions; only the strengthening to the continuous-coefficient ($\star$) retires the regularity-hypothesis requirement.

[1] Cardinality alone does not guarantee a basis element lies in $(0,1)$: rescaling every element of a Hamel basis into $[1,2]$, say, produces a basis that avoids $(0,1)$ entirely. Standard interval-domain constructions often overlook this domain trap; the rational-rescaling device used here is standard (Reem [10, Remark 3.4] uses it, after an idea of Z. Boros, to produce a Hamel basis dense in $\mathbb{R}$) and guarantees $h \in (0,1)$ while keeping basis invariants.

5 Telling $(\star)$ from its relatives: a field guide to saturated Jensen.

Theorem 1 closes the continuous-coefficient equation $(\star)$ with no regularity hypothesis on G; Proposition 7 shows that the same closure fails for the rational-coefficient relative $(J_{\mathbb{Q}})$, where the Hamel pathology survives. A derivation in an applied field that pushes Jensen's inequality to equality therefore lands on one side or the other of a sharp line, and which side it lands on decides whether a regularity hypothesis is doing real work. The purpose of this section is to draw that line precisely and to walk it. The conclusion is not that applied authors habitually carry unnecessary hypotheses; it is subtler and, we think, more useful: the saturated-Jensen identity arises from at least four structurally different sources, only one of which is the regularity-free situation of Theorem 1, and the four are easy to mistake for one another.

5.1 Four sources of a saturated identity, and the questions that separate them.

A saturated Jensen identity

$$p\,G(u_1) + (1-p)\,G(u_2) \;=\; G\big(p\,u_1 + (1-p)\,u_2\big)$$

is, before anything else, a *family* of assertions indexed by the weight p. Its mathematical content is precisely the set of weights for which the surrounding derivation actually licenses the assertion. Four sources recur.

The first is **finite mixing**. If the only primitive in the model is a binary mixture — combine two objects in some proportion — and compound objects are built by finitely iterating it, then the weights one can actually reach are the dyadic rationals, and, after closure under the operation, at most the rationals in $[0,1]$. Such a derivation licenses $(J_{\mathbb{Q}})$, not $(\star)$; by Proposition 7 a regularity hypothesis is then genuinely required, and Theorem 1 does not apply.

The second is **additivity into a bounded codomain**. If the derivation delivers additivity of G together with the information that G takes values in a bounded set, then G is affine by the classical theorem that a bounded additive function is linear — the very results recorded on the two boundedness struts of Figure 1. Here a regularity hypothesis (boundedness) is load-bearing, but it is supplied for free by the codomain rather than assumed by hand, and again the operative mechanism is not the endpoint substitution.

The third is **a Cauchy relative with a non-affine target**. If the equation that actually appears is a relative of Cauchy's equation whose intended solution is *not* affine, then a regularity hypothesis is needed to exclude Hamel-type pathological solutions, and $(\star)$ is simply not the equation in play. The characterization of Shannon entropy is the textbook instance (Section 5.2 below).

The fourth is **a genuine continuum of real weights**. If the weight p is a real parameter realized inside the structure — an atomless measure assigns to events every mass in $[0,1]$, by Sierpiński's intermediate-value theorem for atomless measures [15], of which Lyapunov's convexity theorem [16] is the vector-valued descendant — and the saturated identity is asserted *directly* for that real p, with no detour through finite iteration, then the derivation licenses ($\star$) at every real p, irrationals included. Only now does Theorem 1 apply, and only now is a regularity hypothesis on G vestigial.

This yields an ordered diagnostic. The *zeroth* question is logically prior to the four sources: *does a real-valued solution of the identity provably exist?* Theorem 1 takes a function as input and reports its form; it cannot conjure the function, and an axiom system whose representation theorem fails (the expected-utility example below) fails here, before any weight is examined. Once that gate is cleared, the question that separates the four sources is: *does the derivation assert the identity at every real weight because the weights form a genuine continuum in the structure — or does it merely reach the rational weights, lean on the boundedness of the codomain, or solve a Cauchy relative whose answer is not affine?* In the continuum case the regularity hypotheses of Figure 1 are vestigial and should be dropped; in the others they are load-bearing and must be kept. We do not claim the taxonomy is exhaustive — countable (σ-additive) mixtures, for instance, assert a strictly stronger identity with a theory of its own — but the four sources cover the derivations we have encountered.

The trap, then, is **misclassification**, and it runs in both directions. A derivation of one of the first three kinds can be mistaken for the fourth, so that an author drops a hypothesis that was in fact indispensable. Conversely — and this is the rarer and more seductive error — a derivation of the fourth kind can be mistaken for a Cauchy problem, so that an author *adds* a defensive hypothesis that was never needed. The present paper exists because of an instance of the second error; we return to it in the closing note on provenance (Section 6).

5.2 Where the hypotheses are load-bearing.

We begin on the side of the line where the classical hypotheses earn their keep — which, perhaps against first intuition, is where most of the familiar "saturated Jensen" derivations live.

Expected-utility representation (a continuum that is only apparent). In the von Neumann–Morgenstern tradition [17], sharpened by Herstein and Milnor [18], a preference relation over lotteries that is complete, transitive, independent, mixture-continuous, and Archimedean is represented by a utility functional that is linear in probability,

$$U\big(p\,L_1 + (1-p)\,L_2\big) \;=\; p\,U(L_1) + (1-p)\,U(L_2),$$

and these five conditions are jointly necessary and sufficient (see also [19, 20]). It is tempting to read the displayed identity as ($\star$) and to offer Theorem 1 as a continuity-free substitute for the Archimedean axiom. The temptation should be resisted. The independence axiom is a condition on the *preference relation*; although it is stated for every real mixing weight, it does not by itself deliver a cardinal functional equation for a real-valued U. What produces a real-valued affine U in the first place is the **Archimedean axiom** (which acts mathematically as a topological mixture-continuity condition that embeds the quotient space of indifference classes into the real line), and it is genuinely indispensable: lexicographic preferences are complete, transitive, and independent, yet violate the Archimedean axiom and admit no real-valued representation whatsoever — so there is no U, and the displayed identity, and with it ($\star$), is vacuous. Theorem 1 cannot replace the Archimedean axiom because Theorem 1 *presupposes* a real-valued solution of ($\star$), and the Archimedean axiom is exactly what is needed to produce one. In the language of Section 5.1, this failure is not a deficit of weights — independence is stated at every real p — but a failure at the zeroth gate: there is no real-valued unknown for the equation to constrain. The Archimedean axiom is load-bearing one step *before* the dictionary is even consulted.

The axiomatic characterization of Shannon entropy (a Cauchy relative). In the tradition of Khinchin [21] and Faddeev [22], the entropy $H(p_1, \ldots, p_n) = -\sum_i p_i \log p_i$ is singled out by symmetry, a normalization, and a recursivity (grouping) axiom. The recursivity axiom does not reduce to ($\star$). It reduces to the **fundamental equation of information**, evaluated on the domain $D = \{(x, y) \in [0, 1)^2 \mid x + y \leq 1\}$,

$$f(x) + (1-x)\, f\left(\tfrac{y}{1-x}\right) \;=\; f(y) + (1-y)\, f\left(\tfrac{x}{1-y}\right),$$

for the information function $f(x) = H(x, 1-x)$, and the unique physical solution is $f(x) = -x \log_2 x - (1-x)\log_2(1-x)$, normalized such that $f(0) = f(1) = 0$, which is strictly *concave* — emphatically not affine. The fundamental equation of information is a relative of Cauchy's equation, and like Cauchy's equation it admits Hamel-type non-measurable solutions unless a regularity hypothesis excludes them [23]. This is the decisive point for the present paper, because the hypotheses that have been used to exclude them are *exactly the five struts of Figure 1*. Faddeev assumed continuity of f outright [22] (a clean modern statement is in [24]); the requirement was then weakened, one classical hypothesis at a time — to integrability by Tverberg [25], to measurability by Lee [26], to boundedness on a set of positive measure by Diderrich [27], and to monotonicity on $[0, \frac{1}{2})$ by Kendall [28] and Borges [29]. The entropy characterization is thus the canonical place where the dictionary of Section 3 reappears in an applied setting — and there the hypotheses are *load-bearing*, because the equation is a Cauchy relative with a non-affine target. It is the exact mirror of the calibration computation of Section 5.3, where a saturated identity of the same outward shape is genuinely ($\star$); we record the contrast as a companion to Figure 1 below.

Linear opinion pools (boundedness from the codomain). A third mechanism is worth naming briefly, because it too is easily mistaken for ($\star$). If several experts' probability assignments are aggregated by a rule that commutes with marginalization, the rule must be a weighted average — a linear opinion pool — as soon as the underlying space has at least three points (McConway [30]; Aczél and Wagner [31]). The aggregation function is additive across disjoint events, and it takes values in $[0, 1]$ because its outputs are probabilities; linearity then follows from the bounded-additive theorem of Figure 1, not from the endpoint substitution. As in the utility case, the saturated identity is real, but the work is done by a classical regularity property — here boundedness, supplied by the codomain — rather than by a continuum of weights.

5.3 Where the hypotheses are vestigial.

Now the other side of the line, where the continuum of weights is real and Theorem 1 does its work.

Surrogate calibration on the resolution axis. We first fix the vocabulary, because the central object here travels under many names. Let $(\Omega, \mathcal{F}, \mu)$ be an atomless probability space, let $Y \in \{0,1\}$ be a binary outcome, and let $\mathcal{F}_0 \subseteq \mathcal{F}$ be a sub-σ-algebra of features (or covariates). The posterior probability profile is the $\mathcal{F}_0$-measurable random variable $\eta := \mathbb{E}[Y \mid \mathcal{F}_0] = \Pr(Y = 1 \mid \mathcal{F}_0) \in [0,1]$. For an action space $\mathcal{A}$ and a loss function $\ell\colon \mathcal{A} \times \{0,1\} \to \mathbb{R}$, the *conditional Bayes risk* at posterior value $u \in [0,1]$ is $H(u) := \inf_{a \in \mathcal{A}} \mathbb{E}[\ell(a, Y) \mid \eta = u]$; the function H is variously called the Bayes envelope, the generalized entropy, or the uncertainty function of the associated score [38, 39, 40, 41]. For the 0–1 loss and binary actions $\mathcal{A} = \{0,1\}$, it is the symmetric tent function $H(u) = \min(u, 1-u)$. We write $G\colon [0,1] \to \mathbb{R}$ for a candidate uncertainty function and, for now, let it be *arbitrary*.

A sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}_0$ is a *resolution*: the coarser $\mathcal{G}$, the less the forecast distinguishes states of the world. Writing $\eta_{\mathcal{G}} := \mathbb{E}[\eta \mid \mathcal{G}] = \mathbb{E}[Y \mid \mathcal{G}]$, the risk attainable at resolution $\mathcal{G}$ is

$$U_G(\mathcal{G}) \;:=\; \mathbb{E}\big[G(\eta_{\mathcal{G}})\big], \qquad \text{so that} \quad U_G\big(\sigma(A)\big) \;=\; \mu(A)\,G(u_1) + \mu(A^c)\,G(u_2)$$

when η takes the value u_1 on an event $A \in \mathcal{F}_0$ and u_2 on its complement A^c. The classical theory runs along this axis with concavity in hand. Every Bayes envelope is concave (an infimum of affine functions of u), and for concave G the functional U_G decreases under refinement — conditional Jensen plus the tower property — which is Blackwell's comparison of experiments [42] in this setting and the refinement ordering of DeGroot and Fienberg [39]. The drop it measures is the *resolution* (or discrimination) term in the decomposition of a proper score: Murphy's classical partition of the Brier score [43], Bröcker's generalization to every proper score via its entropy and divergence [44], and the empirical form $\bar{S} = \mathrm{MCB} - \mathrm{DSC} + \mathrm{UNC}$ of Dimitriadis, Gneiting, and Jordan [45]. Every statement in this classical chain *assumes* concavity of G, and the empirical theory adds measurability or smoothness besides.

The dictionary's contribution is the converse direction, and it needs none of that.

Proposition 8 (Resolution-blind uncertainty functions are affine)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be atomless and let $G\colon [0,1] \to \mathbb{R}$ be arbitrary. Suppose the two-cell resolutions are exact: for all $u_1, u_2 \in [0,1]$ and every event A,*

$$\mu(A)\,G(u_1) + \mu(A^c)\,G(u_2) \;=\; G\big(\mathbb{E}[\eta]\big), \qquad \textit{where } \eta = u_1 \mathbf{1}_A + u_2 \mathbf{1}_{A^c}.$$

Then G is affine; consequently $U_G(\mathcal{G}) = G(\mathbb{E}[\eta])$ for every *$[0,1]$-valued posterior η and* every *sub-σ-algebra $\mathcal{G}$, so G is blind to resolution at all scales. No continuity, measurability, monotonicity, boundedness — or concavity — is assumed. The k-class case, with η valued in the probability simplex Δ_k and G arbitrary on Δ_k, holds verbatim with "affine on the simplex," by Theorem 6.*

Proof. Atomlessness realizes every mass $p = \mu(A) \in [0,1]$ (Sierpiński [15]), and the values u_1, u_2 are prescribed independently of A, so the hypothesis asserts $p\,G(u_1) + (1-p)\,G(u_2) = G(p\,u_1 + (1-p)\,u_2)$ at every triple — the identity ($\star$) on its entire domain, with G only ever *evaluated* at points, never integrated. Theorem 1 gives affineness on $[0,1]$; Theorem 6 with $C = \Delta_k$ gives the simplex case. Conversely, for affine G and any $\mathcal{G}$, linearity of expectation and the tower property give $\mathbb{E}[G(\eta_{\mathcal{G}})] = G(\mathbb{E}[\eta_{\mathcal{G}}]) = G(\mathbb{E}[\eta])$. □

This is the seductive misclassification of Section 5.1 in the flesh: the saturated identity looks like a Cauchy problem and invites a defensive boundedness or measurability hypothesis on G, but the continuum of cell masses makes it genuine ($\star$) and the hypothesis vestigial.

The benchmark partition and the tent. What survives when exactness holds only *locally* is Corollary 5: a surrogate that aggregates exactly on each cell of a fixed benchmark partition of the probability axis is affine on each cell, with continuity at the knots for free. For the canonical benchmark $\{[0, \frac{1}{2}], [\frac{1}{2}, 1]\}$ the resulting shape is the tent — and the tent is not merely an example. $T_{1/2}(u) = \min(u, 1-u)$ *is* the 0–1 conditional Bayes risk, and more is true: the elementary scores of Schervish [46] and of Ehm, Gneiting, Jordan, and Krüger [47] have as their uncertainty functions precisely the tents $T_\theta(u) := 2\min\big(u(1-\theta),\,(1-u)\theta\big)$ with apex at $\theta \in (0,1)$, and the mixture representation $S = \int_0^1 S_\theta\, dH(\theta)$ of every proper score exhibits every admissible uncertainty function as a mixture of tents, $G = \int_0^1 T_\theta\, dH(\theta)$ — for instance $\int_0^1 T_\theta\, d\theta = u(1-u)$, the Brier entropy. The tents are the extreme rays of the cone of uncertainty functions, and piecewise saturation pins exactly them. In learning-theoretic language, cellwise exactness is tightness of the surrogate-risk bracket: the ψ-transform of Bartlett, Jordan, and McAuliffe [32], Steinwart's calibration functions [48], and the multiclass analysis of Tewari and Bartlett [49] measure how far a surrogate's floor sits from the true floor; exactness on a cell says the gap is zero there, and Corollary 5 says the price of zero gap is affineness on the cell. Figure 3 draws both halves.

Two readings. The construction travels. In probabilistic graphical models, a clique tree is *calibrated* when neighboring cliques agree on their sepset marginals [50]; this cellwise agreement is the piecewise identity above, asserted over the sepset partition, and U_G is the natural uncertainty functional along the junction tree's resolution order. In representation learning, an encoder R induces the resolution $\mathcal{G} = \sigma(R)$, and $U_{T_{1/2}}(\sigma(R))$ is the smallest misclassification error achievable from the representation — its error floor. Proposition 8 then says that the only uncertainty functions blind to the choice of representation are the affine ones: any genuinely concave G certifies, through a strict drop of U_G, that the representation gained resolution.

The thin line, in miniature. Replace the atomless space by a single fair coin — build the mixtures by tossing it repeatedly — and the attainable masses p collapse to the dyadic rationals; one is back in the first source of Section 5.1, Proposition 7 returns a

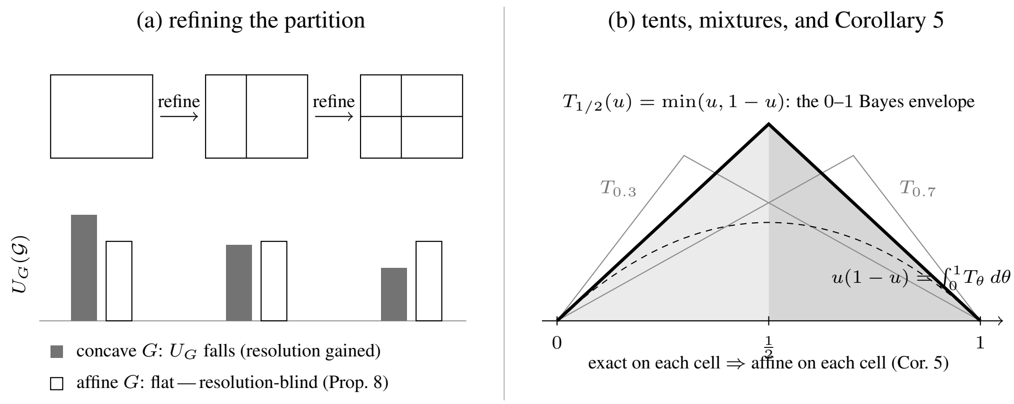

Figure 3: Saturation along the resolution axis. (a) Refining the partition can only lower the uncertainty functional $U_G(\mathcal{G}) = \mathbb{E}[G(\eta_{\mathcal{G}})]$ when G is concave (conditional Jensen; Blackwell [42]); the drop is the resolution term of the score decompositions [43, 44, 45]. By Proposition 8, the only uncertainty functions whose profile is flat — resolution-blind at every two-cell comparison — are the affine ones, with no regularity assumed. (b) The tent $T_{1/2}(u) = \min(u, 1-u)$ is the 0–1 Bayes envelope; the tents T_θ are the uncertainty functions of the elementary scores [46, 47], and every admissible uncertainty function is a mixture $\int_0^1 T_\theta \, dH(\theta)$; the dashed curve is the Brier entropy $u(1-u)$, the uniform mixture. Exactness on each cell of the benchmark partition $\{[0, \frac{1}{2}], [\frac{1}{2}, 1]\}$ forces affineness on each cell (Corollary 5): the tent shape, pinned with nothing assumed at the kink.

non-affine solution, and a regularity hypothesis is once more required.[2] The whole of Sections 3 and 5 is visible in the gap between an atomless measure and a coin.

Mixture-affine assignments on quantum states. A second instance of the fourth source — this one consuming the higher-dimensional Theorem 6 — arises in the foundations of quantum mechanics. The density operators $\mathcal{S}_d$ on $\mathbb{C}^d$ form a convex set, and the statistical mixture $p\,\rho_1 + (1-p)\,\rho_2$ is operationally primitive: prepare ρ_1 with probability p using a classical randomizer, with p ranging over a genuine continuum. Let $v\colon \mathcal{S}_d \to \mathbb{R}$ be any assignment respecting mixtures,

$$v\big(p\,\rho_1 + (1-p)\,\rho_2\big) \;=\; p\,v(\rho_1) + (1-p)\,v(\rho_2) \qquad (\rho_1, \rho_2 \in \mathcal{S}_d,\; p \in [0,1]).$$

This is exactly the hypothesis of Theorem 6 on the convex set $\mathcal{S}_d$, so v extends to a linear-plus-constant functional; since every linear functional on the finite-dimensional real space of Hermitian operators has the form $u \mapsto \operatorname{tr}(Bu)$, we get $v(\rho) = \operatorname{tr}(A\rho) + b$ for a Hermitian A — with no positivity, boundedness, continuity, or measurability assumed of v. The affine representation itself is operational folklore (see, e.g., Holevo [33,

[2] A terminological caution: the property tested throughout is that G *commutes with expectation* over two-point laws — we say the family is *expectation-exact* — rather than "mean-preserving," a term that already names mean-preserving spreads in economics.

Lemma 1.6.1]); what the dictionary adds is a diagnosis of why the dual derivations must pay more. Gleason-type theorems work on the *measurement* side: a frame function on effects is additive over coarse-grainings of measurements (where grouping disjoint measurement outcomes corresponds to operator addition, $E_1 + E_2 \leq I$), and coarse-graining is finite/rational mixing — the first source. Additivity yields homogeneity only over the nonnegative rationals, and the rational-to-real step must then be purchased with a regularity hypothesis: nonnegativity in Gleason's original projection-based theorem [34] (for Hilbert space dimensions $d \geq 3$) and in Busch's POVM effect-algebra version [35] (which holds for all $d \geq 2$), positivity or an explicit continuity proof in Caves, Fuchs, Manne, and Renes [36], and, in the Cauchy-equation treatment of Wright and Weigert [37], the catalog that nonlinear, pathological frame functions based on a Hamel basis can be bounded on neither side, continuous at zero, nor Lebesgue measurable — the catalog of Figure 1 rediscovered inside physics. State preparation tests the identity directly at every real weight $p \in [0, 1]$ via classical analog mixing; measurement coarse-graining tests it only additivity-wise, allowing dense Hamel-basis oscillations unless bounded or positive. The dictionary predicts exactly where each route must spend a regularity hypothesis, and the derivations cited above spend it exactly there.

5.4 The dictionary, mirrored.

Figure 1 records, for a *single* equation, which hypotheses force affineness. The diagnostic of this section is that the *same five hypotheses* sit on opposite sides depending on the source of the identity. Table 2 places the load-bearing case of Section 5.2 (the entropy characterization, a Cauchy relative) beside the vestigial case of Section 5.3 (the atomless calibration computation, genuine ($\star$)); the contrast between its two columns is the content of the section.

5.5 The lesson.

The recurrence worth articulating is not a habit of carrying unnecessary hypotheses; it is the ease of misreading which of the four sources one is in. Saturated Jensen is produced by finite mixing, by additivity into a bounded codomain, by Cauchy relatives with non-affine targets, and by genuine continua of real weights, and only the last retires the regularity hypotheses. A zeroth gate precedes them all: absent an existence axiom there is no real-valued unknown, and the question of weights never arises. The diagnostic of Section 5.1 is offered as the means of telling them apart; the mirrored dictionary of Section 5.4 is the warning that the same five classical hypotheses appear on both sides. An author who has internalized the Cauchy–Hamel toolkit is well defended against the first error and peculiarly exposed to the second — which is, as the closing note on provenance recounts, how this paper began.

Hypothesis on the unknown function	**In the entropy characterization (a Cauchy relative)**	**Under a saturated ($\star$) on an atomless space**
Continuity	**Required** (Faddeev [22]).	**Vestigial** — a conclusion (Corollary 2).
Measurability	**Required** (Lee [26]).	**Vestigial** — same.
Monotonicity	**Required** (Kendall [28], Borges [29]).	**Vestigial** — same.
Boundedness on a set of positive measure	**Required** (Diderrich [27]).	**Vestigial** — same.
Boundedness on the interval	**Required** (a fortiori).	**Vestigial** — same.
None	**Insufficient** — Hamel-type solutions of the fundamental equation of information [23].	**Sufficient** — Theorem 1.

Table 2: The dictionary of Figure 1, mirrored. The same five hypotheses are indispensable when the operative equation is the fundamental equation of information (a relative of Cauchy's equation, whose solution is the concave Shannon function), and vestigial when a genuine continuum of weights makes the operative equation the continuous-coefficient ($\star$).

6 Concluding remarks.

Theorem 1 is, mathematically, one line. The dictionary of Section 3 is a table. The structural argument of Section 3.2 is one sentence: *the Hamel pathology lives at irrational p; the continuous-coefficient equation* ($\star$) *forecloses on it there.* Why does this material deserve an article in the MONTHLY?

Two reasons. First, the classical detour through Hamel and the Polish school produced *foundational* results of classical analysis — and the dictionary of Section 3 articulates a precise sense in which the detour answered an adjacent question: not "what is the minimum hypothesis for ($\star$)⇒affine?" (Theorem 1's answer: nothing), but "what is the minimum hypothesis for (J_2)⇒affine?" (Polish school's answer: any of a dictionary of tameness conditions). Both questions are interesting; their answers are different.

Second, the diagnostic of Section 5 is, to our knowledge, not articulated explicitly elsewhere in the functional-equations literature. The same five regularity hypotheses turn out to be indispensable in some saturated-Jensen settings (the Shannon-entropy characterization) and vestigial in others (a calibration computation on an atomless space), and the explicit criterion may save other authors from misreading which case they are in — in either direction. Nor is the asymmetry hypothetical: in the Gleason-type derivations of the quantum Born rule, the measurement-side route must purchase the rational-to-real step with positivity or one of the dictionary's hypotheses [34, 35, 36, 37], while the preparation-side route of Section 5.3 pays nothing — the two columns of Table 2, alive in a single literature.

The endpoint substitution itself is the kind of one-line proof that, once seen, is hard to unsee — but that nonetheless takes some scaffolding to make visible. The present paper is the scaffolding.

A note on provenance.

The present observation reached the author through an unconventional route: a *formalization project* on a separate topic, in which the lemma "G satisfies ($\star$) on $[0, M]$ implies G is affine" was initially stated with a defensive boundedness hypothesis on G, in deference to the Cauchy–Hamel literature. When the formal proof was written out, the boundedness hypothesis turned out to be unused: the endpoint substitution closed the proof on its own. A literature search then confirmed that the substitution technique is folklore: Aczél's [11, §2.1.4] proves a dyadic precursor under a continuity hypothesis. The precise continuous-coefficient regularity-free statement of Theorem 1, while elementary, does not appear in this exact form in the standard references the author consulted; the present paper grew out of the attempt to explain, to the author's own satisfaction, why the boundedness hypothesis had felt necessary in the first place, and to articulate the contrast between the discrete- and continuous-coefficient cases as a dictionary. The lesson is general: a defensive regularity hypothesis is not free, and a formalization project occasionally surfaces such hypotheses precisely because the formal system requires every clause of the statement to be used.

Acknowledgments.

AI Assistance Disclosure. A proprietary neuro-symbolic AI research harness incorporating adversarial search through Program Synthesis was used during exploratory theorem-discovery and formalization activities. The harness employed Claude Opus 3.7 and Lean 4, and was behind the initial observation, denoted ($\star$), that motivated subsequent investigation. The development, interpretation, mathematical analysis, and presentation of the results were carried out by the author. Claude Fable 5 was additionally used for proofreading, editorial review, and formatting assistance. The author takes full responsibility for the correctness, originality, and content of the manuscript.

Disclosure statement.

The author reports there are no competing interests to declare.

Funding.

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

References

[1] Hamel G. Eine Basis aller Zahlen und die unstetigen Lösungen der Funktionalgleichung $f(x+y) = f(x) + f(y)$. Math Ann. 1905;60:459-62.

[2] Cauchy AL. Cours d'analyse de l'École royale polytechnique. Première partie: Analyse algébrique. Paris: Imprimerie royale; 1821.

[3] Darboux G. Sur la composition des forces en statique. Bull Sci Math Astron. 1875;9:281-8.

[4] Darboux G. Sur le théorème fondamental de la géométrie projective. Math Ann. 1880;17:55-61.

[5] Sierpiński W. Sur les fonctions convexes mesurables. Fund Math. 1920;1:125-9.

[6] Kormes M. On the functional equation $f(x+y) = f(x)+f(y)$. Bull Amer Math Soc. 1926;32:689-93.

[7] Ostrowski AM. Über die Funktionalgleichung der Exponentialfunktion und verwandte Funktionalgleichungen. Jahresber Deutsch Math-Verein. 1929;38:54-62.

[8] Kestelman H. On the functional equation $f(x + y) = f(x) + f(y)$. Fund Math. 1947;34:144-7.

[9] Steinhaus H. Sur les distances des points dans les ensembles de mesure positive. Fund Math. 1920;1:93-104.

[10] Reem D. Remarks on the Cauchy functional equation and variations of it. Aequationes Math. 2017;91. doi:10.1007/s00010-016-0463-6.

[11] Aczél J. Lectures on Functional Equations and Their Applications. vol. 19 of Mathematics in Science and Engineering. New York: Academic Press; 1966.

[12] Kuczma M. An Introduction to the Theory of Functional Equations and Inequalities. Cauchy's Equation and Jensen's Inequality. 2nd ed. Basel: Birkhäuser; 2009. Second edition edited by A. Gilányi.

[13] Aumann RJ. Markets with a continuum of traders. Econometrica. 1964;32(1-2):39-50.

[14] Aczél J, Dhombres J. Functional Equations in Several Variables. vol. 31 of Encyclopedia of Mathematics and its Applications. Cambridge: Cambridge University Press; 1989.

[15] Sierpiński W. Sur les fonctions d'ensemble additives et continues. Fund Math. 1922;3:240-6.

[16] Lyapunov AA. Sur les fonctions-vecteurs complètement additives. Izv Akad Nauk SSSR Ser Mat. 1940;4:465-78. Accessible modern exposition: Halmos, P. R., "The range of a vector measure," *Bull. Amer. Math. Soc.* 54 (1948) 416–421.

[17] von Neumann J, Morgenstern O. Theory of Games and Economic Behavior. Princeton: Princeton University Press; 1944.

[18] Herstein IN, Milnor J. An axiomatic approach to measurable utility. Econometrica. 1953;21:291-7.

[19] Fishburn PC. Utility Theory for Decision Making. New York: Wiley; 1970.

[20] Kreps DM. Notes on the Theory of Choice. Underground Classics in Economics. Boulder, CO: Westview Press; 1988.

[21] Khinchin AI. Mathematical Foundations of Information Theory. New York: Dover; 1957.

[22] Faddeev DK. Zum Begriff der Entropie einer endlichen Wahrscheinlichkeitsschemas. In: Arbeiten zur Informationstheorie. vol. 1. Berlin: Deutscher Verlag der Wissenschaften; 1957. p. 85-90.

[23] Aczél J, Daróczy Z. On Measures of Information and Their Characterizations. vol. 115 of Mathematics in Science and Engineering. New York: Academic Press; 1975.

[24] Baez JC, Fritz T, Leinster T. A characterization of entropy in terms of information loss. Entropy. 2011;13(11):1945-57.

[25] Tverberg H. A new derivation of the information function. Math Scand. 1958;6:297-8.

[26] Lee PM. On the axioms of information theory. Ann Math Statist. 1964;35:415-8.

[27] Diderrich GT. The role of boundedness in characterizing Shannon entropy. Inform and Control. 1975;29(2):149-61.

[28] Kendall DG. Functional equations in information theory. Z Wahrsch Verw Gebiete. 1964;2:225-9.

[29] Borges R. Zur Herleitung der Shannonschen Information. Math Z. 1967;96:282-7.

[30] McConway KJ. Marginalization and linear opinion pools. J Amer Statist Assoc. 1981;76(374):410-4.

[31] Aczél J, Wagner C. A characterization of weighted arithmetic means. SIAM J Algebraic Discrete Methods. 1980;1(3):259-60.

[32] Bartlett PL, Jordan MI, McAuliffe JD. Convexity, classification, and risk bounds. J Amer Statist Assoc. 2006;101(473):138-56.

[33] Holevo AS. Probabilistic and Statistical Aspects of Quantum Theory. vol. 1 of North-Holland Series in Statistics and Probability. Amsterdam: North-Holland; 1982.

[34] Gleason AM. Measures on the closed subspaces of a Hilbert space. J Math Mech. 1957;6:885-93.

[35] Busch P. Quantum states and generalized observables: a simple proof of Gleason's theorem. Phys Rev Lett. 2003;91:120403.

[36] Caves CM, Fuchs CA, Manne KK, Renes JM. Gleason-type derivations of the quantum probability rule for generalized measurements. Found Phys. 2004;34(2):193-209.

[37] Wright VJ, Weigert S. Gleason-type theorems from Cauchy's functional equation. Found Phys. 2019;49. doi:10.1007/s10701-019-00275-x.

[38] Savage LJ. Elicitation of personal probabilities and expectations. J Amer Statist Assoc. 1971;66(336):783-801.

[39] DeGroot MH, Fienberg SE. The comparison and evaluation of forecasters. J Roy Statist Soc Ser D (The Statistician). 1983;32(1–2):12-22.

[40] Grünwald PD, Dawid AP. Game theory, maximum entropy, minimum discrepancy and robust Bayesian decision theory. Ann Statist. 2004;32(4):1367-433.

[41] Gneiting T, Raftery AE. Strictly proper scoring rules, prediction, and estimation. J Amer Statist Assoc. 2007;102(477):359-78.

[42] Blackwell D. Equivalent comparisons of experiments. Ann Math Statist. 1953;24:265-72.

[43] Murphy AH. A new vector partition of the probability score. J Appl Meteorol. 1973;12:595-600.

[44] Bröcker J. Reliability, sufficiency, and the decomposition of proper scores. Q J R Meteorol Soc. 2009;135(643):1512-9.

[45] Dimitriadis T, Gneiting T, Jordan AI. Stable reliability diagrams for probabilistic classifiers. Proc Natl Acad Sci USA. 2021;118(8):e2016191118.

[46] Schervish MJ. A general method for comparing probability assessors. Ann Statist. 1989;17(4):1856-79.

[47] Ehm W, Gneiting T, Jordan A, Krüger F. Of quantiles and expectiles: consistent scoring functions, Choquet representations, and forecast rankings. J R Stat Soc Ser B. 2016;78(3):505-62.

[48] Steinwart I. How to compare different loss functions and their risks. Constr Approx. 2007;26(2):225-87.

[49] Tewari A, Bartlett PL. On the consistency of multiclass classification methods. J Mach Learn Res. 2007;8:1007-25.

[50] Koller D, Friedman N. Probabilistic Graphical Models: Principles and Techniques. Cambridge, MA: MIT Press; 2009.

[51] Solovay RM. A model of set-theory in which every set of reals is Lebesgue measurable. Ann of Math (2). 1970;92(1):1–56.